

Metabolite Interaction Analysis 2

Donguk Shin

Seoul National University
Department of Statistics

Main Objectives and Outline

Main Objectives

- To investigate whether interactions among plant metabolites influence insect performance.
- To simplify the existing analysis.

- Introduction
- Analysis Results
- Discussion

- The dataset consists of:
 - Insect performance (response, Y).
 - Metabolite abundances (covariates, X).
 - Four groups (1e, mp, px, sl).
- Each group contains 24 observations, resulting in a total of 96 samples ($n = 96$).
- X comprises 4,904 variables ($p = 4904$), including a control group.

- Because the response is positive, a log-transformation was applied:

$$y_i \leftarrow \log(y_i).$$

- Using the control group, metabolite abundances were transformed into log2FC:

$$x_{ij} \leftarrow \log_2 x_{ij} - \log_2 x'_i,$$

where x'_i denotes the corresponding value from the control group.

- Subsequently, group-wise normalization was performed.

Previous Analysis

Feature Screening:

1. HSIC-ordering (Song et al., 2007).
2. HSIC-Lasso (Yamada et al., 2014).

Model Selection:

1. Penalized linear regression approach: hierNet (Bien et al., 2013).
2. Penalized quadratic linear regression approach: sprinter (Yu et al., 2020).

Limitation:

- The screening and model-selection procedures were separate.
 - A model-selection method that does not require prior screening is needed.
 - A forward-selection approach that adds variables sequentially from the null model is appropriate.

iForm: Interaction-selection procedure featuring FORward selection (Hao and Zhang, 2014).

- Assumes a quadratic linear model.
- Assumes weak heredity.
- Incrementally adds main effects and possible interactions one at a time (forward selection).
- Selects the final model using the Bayesian Information Criterion (BIC).
- This possesses the sure screening property in an ultra-high-dimensional setting.

Methods – Stage-wise Regression

RAMP: Regularization algorithm under the marginality principle (Zhang et al., 2016).

- Assumes a quadratic linear model.
- Assumes weak heredity.
- Updates the solution path starting of the penalized regression from the null model.
- Implements a stage-wise regression approach, similar to forward selection.
- Selects the final model using the extended Bayesian Information Criterion (EBIC).
- Uses the SCAD penalty (Fan and Li, 2001).

Results – Group 1e

Interaction Detection:

- **iForm:** No interactions detected.
- **RAMP:** $X_{747} : X_{3004}$, $X_{762} : X_{3004}$, $X_{559} : X_{1797}$, and $X_{2506} : X_{3004}$.

Conclusion: For Group 1e, many interactions were chosen by RAMP.

Interaction Detection:

- **iForm:** No interactions detected.
- **RAMP:** No interactions detected.

Conclusion: For Group mp, no interactions were detected.

Interaction Detection:

- **iForm:** No interactions detected.
- **RAMP:** $X_{133} : X_{1114}$, $X_{133} : X_{1182}$, $X_{133} : X_{1216}$, $X_{266} : X_{1755}$, $X_{266} : X_{1898}$.

Conclusion: For Group px, many interactions were chosen by RAMP.

Results – Group s1

Interaction Detection:

- **iForm:** No interactions detected.
- **RAMP:** No interactions detected.

Conclusion: For Group s1, no interactions were detected.

Conclusions:

- For Groups 1e and px, many interactions were selected by RAMP.
- Consistent finding: certain groups exhibit a large number of detected interactions.

Limitations:

- Limited reproducibility: the set of selected interactions varies across methods.
- Only model selection: precise hypothesis testing within the model is not supported.

Suggestions:

- The sample size is extremely limited; additional data collection is necessary.
- Because some covariates show strong inter-correlation, clustering of covariates appears necessary.
- Propose a procedure for covariate clustering, low-dimensional representation of clusters, and exploration of interactions among clusters.

Reference

- Song, L., Smola, A., Gretton, A., Bedo, J., & Borgwardt, K. (2007). Supervised feature selection via dependence estimation. ICML 2007.
- Yamada, M., Jitkrittum, W., Sigal, L., Xing, E. P., & Sugiyama, M. (2014). High-dimensional feature selection by feature-wise kernelized Lasso. Neural Information Processing Systems (NeurIPS 2014).
- Bien, J., Taylor, J., & Tibshirani, R. (2013). A lasso for hierarchical interactions. *Annals of Statistics*, 41(3), 1111–1141.
- Yu, Guo, Jacob Bien, and Ryan Tibshirani. "Reluctant interaction modeling." arXiv preprint arXiv:1907.08414 (2019).
- Hao, Ning, and Hao Helen Zhang. "Interaction screening for ultrahigh-dimensional data." *Journal of the American Statistical Association* 109.507 (2014): 1285-1301.
- Hao, Ning, Yang Feng, and Hao Helen Zhang. "Model selection for high-dimensional quadratic regression via regularization." *Journal of the American Statistical Association* 113.522 (2018): 615-625.
- Fan, Jianqing, and Runze Li. "Variable selection via nonconcave penalized likelihood and its oracle properties." *Journal of the American statistical Association* 96.456 (2001): 1348-1360.